THE MASTERMIND COACHING CLASSES

Revision Notes

Class 10 · Mathematics

Chapters 10 to 14 · NCERT Mathematics — Class 10 · CBSE / RBSE Syllabus 2026-27

HOW TO USE THESE NOTES

Each chapter follows the textbook's own order of sections, so you can read straight through. Bold words are keywords you must be able to define. Tables compare and classify; shaded boxes hold definitions, formulas and key facts. The navy Key Takeaways box at the end of every chapter is your last-two-minutes revision.

CH.	CHAPTER TITLE	CORE THEME
10	Circles	Tangent to a circle, perpendicularity to the radius, equal tangent lengths
11	Areas Related to Circles	Circumference and area, sector, segment, combinations of plane figures
12	Surface Areas and Volumes	Surface area and volume of combinations of solids, conversion of one solid into another
13	Statistics	Mean, median and mode of grouped data
14	Probability	Theoretical probability, complementary events, dice, coins and cards

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CHAPTER 10 · MATHS

Circles

A wheel rolling on a road touches it at just one point at a time, a pulley belt leaves the wheel along a straight line, and a ball resting on the floor meets it at a single point. These are everyday pictures of a tangent to a circle. This chapter studies the lines that meet a circle at one point or two, proves two key theorems about tangents and uses them to solve problems.

Big Questions

- How can a line meet a circle: in no point, one point or two?
- Why is a tangent perpendicular to the radius at the point of contact?
- Why are the two tangents from an outside point equal in length?

10.1 Tangent and Secant

Line	Points common with the circle	Name
Does not touch	0	Non-intersecting line
Touches	**1** (point of contact)	**Tangent**
Cuts	**2**	**Secant** (the part inside is a chord)

DEFINITION

A **tangent** to a circle is a line that **intersects the circle at exactly one point**, called the **point of contact**. A secant becomes a tangent when its two intersection points coincide.

- At any point on a circle there is **one and only one** tangent.
- From a point **inside** the circle no tangent can be drawn; from a point **on** the circle there is one; from a point **outside** there are **exactly two** tangents.
- A circle can have **any number of** parallel tangents, but only two parallel to a given direction (at the ends of a diameter).

10.2 Tangent is Perpendicular to the Radius

DEFINITION

Theorem 1: The tangent at any point of a circle is **perpendicular** to the radius through the point of contact.

Idea of the proof. Let the tangent XY touch the circle with centre O at P, and take any other point Q on XY. Q lies outside the circle, so OQ is longer than the radius OP. Thus OP is the **shortest distance** from O to the line XY, and the shortest distance is the perpendicular. Hence OP is perpendicular to XY.

REMEMBER

Corollary: the **normal** at the point of contact (the line perpendicular to the tangent) passes through the centre. So the perpendicular to a tangent at the point of contact passes through the centre.

10.3 Lengths of Tangents from an External Point

DEFINITION

Theorem 2: The lengths of the two tangents drawn to a circle from an **external point** are **equal**.

Proof. Let PA and PB be tangents from P to a circle with centre O. In right triangles OAP and OBP: OA = OB (radii), OP is common, angle OAP = angle OBP = 90°. So the triangles are congruent (RHS), and **PA = PB**. It also follows that angle APO = angle BPO, so **OP bisects the angle between the tangents**, and angle OPA = angle OPB.

Result	Statement
Angle relation	Angle between two tangents + angle subtended by the points of contact at the centre = **180°**
Isosceles	Triangle PAB is isosceles: angle PAB = angle PBA = 90° − (angle APB)/2
Perpendicular bisector	OP is the perpendicular bisector of the chord of contact AB
Quadrilateral	A quadrilateral circumscribing a circle has **AB + CD = AD + BC**

10.4 Worked Examples

Length of a tangent. OP = 13 cm from the centre with radius 5 cm: OT is perpendicular to PT, so PT = $\sqrt{(13^2 - 5^2)} = \sqrt{144} =$ **12 cm**. The tangent length from a point at distance d is $\sqrt{(d^2 - r^2)}$.

Concentric circles. Radii 5 cm and 3 cm; a chord of the bigger circle touches the smaller. The radius of the smaller circle to the point of contact is perpendicular to the chord and bisects it: half-chord = $\sqrt{(25 - 9)} = 4$, so the chord is **8 cm**.

Angle between tangents. If the angle between two tangents is 60°, angle AOB = 180° − 60° = **120°**; each of angles OPA and OPB is 30°. If two tangents PA and PB from P make angle 80°, angle PAB = 50°.

Quadrilateral. A quadrilateral ABCD circumscribes a circle with AB = 6, CD = 4, BC = 7: AB + CD = AD + BC gives 10 = AD + 7, so **AD = 3**.

Prove that the tangents drawn at the ends of a diameter are parallel: each is perpendicular to the diameter (a radius at each end), so they are parallel (the interior angles sum to 180°).

Prove that the parallelogram circumscribing a circle is a rhombus: opposite sides equal gives AB + CD = AD + BC, so 2AB = 2BC, i.e. all sides equal.

IMPORTANT

Always join the centre to the point of contact first; the right angle you get lets you apply Pythagoras or congruence. In a circle diagram, tangents from one point are equal, so mark equal lengths.

Quick Revision: Likely Exam Questions

Likely question	Model answer
How many tangents can be drawn from an external point?	Two.

Likely question	Model answer
What is the angle between a tangent and the radius at the point of contact?	90°.
PT is a tangent from P, OP = 25 cm, radius 7 cm. Find PT.	24 cm.
What is the length of a tangent from a point at distance 10 from the centre of a circle of radius 6?	8 cm.
Why are the tangents from an external point equal?	Triangles OAP and OBP are congruent by RHS.
Define a secant.	A line that intersects the circle at two points.
If angle AOB = 110°, find the angle between tangents at A and B.	70°.
State a property of a circumscribed quadrilateral.	Sum of one pair of opposite sides equals the sum of the other pair.

★ KEY TAKEAWAYS

- Tangent meets the circle at **one point**; secant at **two**.
- **Tangent ⊥ radius** at the point of contact.
- **Equal tangents** from an external point; OP bisects the angle between them.
- Angle between tangents + angle at centre = **180°**; tangent length = $\sqrt{(d^2 - r^2)}$.

CHAPTER 11 · MATHS

Areas Related to Circles

A pizza slice, the sweep of a windscreen wiper, a lawn watered by a rotating sprinkler and a running track all involve parts of circles. This chapter recalls the circumference and area of a circle and extends them to sectors and segments, then uses them for areas of shapes made by combining squares, triangles and circles.

Big Questions

- How are the area and length of a sector or arc related to the whole circle?
- How is the area of a segment found?
- How do we find areas of combined or shaded figures?

11.1 Circle: Revision

FORMULA

Circumference = $2\pi r = \pi d$. Area = πr^2. Use $\pi = 22/7$ when the radius is a multiple of 7, otherwise 3.14, unless stated. **Semicircle:** area $\pi r^2/2$, perimeter $\pi r + 2r$. **Ring (annulus)** between radii R and r: area = $\pi(R^2 - r^2)$.

Examples. A wheel of diameter 70 cm has circumference 220 cm, so it makes 110000/220 = **500 revolutions** in 1.1 km. A ring between circles of radii 14 cm and 7 cm has area (22/7)(196 − 49) = **462 cm^2**. If the circumference is 44 cm, then r = 7 cm and area = 154 cm^2.

11.2 Sector and Arc

DEFINITION

A **sector** is the region bounded by two radii and the arc between them. The smaller one is the **minor sector** and the larger the **major sector** (angle θ and 360° − θ).

FORMULA

For a sector of angle **θ (in degrees)** and radius r: **Length of arc = (θ/360) x 2πr. Area of sector = (θ/360) x πr^2 = (1/2) x arc length x r. Perimeter of sector = 2r + arc length.**

Radius	Angle	Arc length	Area of sector
6 cm	60°	2π = 6.28 cm	6π = 18.86 cm^2
14 cm	90°	22 cm	154 cm^2
5 cm	72°	2π = 6.28 cm	5π = 15.7 cm^2
21 cm	120°	44 cm	462 cm^2

Clock example. The minute hand of a clock is 14 cm long. In 5 minutes it turns 30°, so the area swept is (30/360)(22/7)$(14)^2$ = **154/3 = 51.33 cm^2**; the tip travels 22/3 = 7.33 cm. In 1 hour the tip travels 2πr = 88 cm.

11.3 Segment of a Circle

DEFINITION

A **segment** is the region between a chord and the arc. The smaller is the **minor segment**; the other is the **major segment**.

FORMULA

Area of minor segment = area of sector − area of triangle OAB = $(\theta/360)\pi r^2 - (1/2)r^2 \sin\theta$. **Area of major segment = πr^2 − area of minor segment.** For $\theta = 90°$, the triangle has area $r^2/2$; for $\theta = 60°$ it is equilateral with area $(\sqrt{3}/4)r^2$; for $\theta = 120°$ the triangle area is $(\sqrt{3}/4)r^2$ too.

Example. A chord subtends 90° at the centre of a circle of radius 10 cm. Sector = (1/4)(3.14)(100) = 78.5; triangle = (1/2)(10)(10) = 50; **minor segment = 28.5 cm^2**, major segment = 314 − 28.5 = 285.5 cm^2. **60° segment of radius 6 cm:** sector 6π = 18.84; triangle (√3/4) x 36 = 15.57; segment = **3.27 cm^2**.

11.4 Combinations of Plane Figures

Step 1

Break the figure into squares, rectangles, triangles, sectors and segments.

Step 2

Add or subtract the areas depending on whether the part is included or removed (shaded region = outer figure − unshaded parts).

Step 3

Identify radii and angles from the figure, particularly equal arcs and quarter-circles.

Figure	Area / perimeter
Square of side 14 cm with four quarter circles of radius 7 cm at its corners removed	196 − 154 = **42 cm^2**
Running track: rectangle with a semicircle at each end	rectangle + πr^2; perimeter = $2l + 2\pi r$
Circle inscribed in a square of side a	Area = $\pi a^2/4$, shaded corners = $a^2(1 - \pi/4)$
Square inscribed in a circle of radius r	Side = $r\sqrt{2}$; area of square $2r^2$; circle left over = $(\pi - 2)r^2$

Example. A horse is tied with a 5 m rope at a corner of a square field: it can graze the quarter circle (1/4)(3.14)(25) = **19.63 m^2**. If the rope is 10 m, the quarter circle gives 78.5 m^2.

IMPORTANT

Use the angle in **degrees** in the sector formulas. When the question says 'minor' or 'major', check the angle; if the angle of the sector is not given, find it from the chord using triangle properties.

Quick Revision: Likely Exam Questions

Likely question	Model answer
Write the area of a sector of angle θ.	$(\theta/360) \times \pi r^2$.

Likely question	Model answer
Length of the arc of a sector of angle 60° and radius 21 cm?	22 cm.
What is a segment?	The region between a chord and the arc.
Area of a ring between radii 7 and 5?	$\pi(49 - 25) = 24\pi = 75.43$ cm^2 (with $\pi = 22/7$).
Perimeter of a quadrant of radius 7?	11 + 14 = 25 cm.
How is a minor segment's area found?	Sector area minus triangle area.
How many times does a wheel of radius 35 cm revolve in 22 m?	22/2.2 = 10.
Area of a semicircle with diameter 14 cm?	77 cm^2.

★ KEY TAKEAWAYS

- **Arc = (θ/360) 2πr; sector = (θ/360) πr^2.**
- **Minor segment = sector − triangle**; major segment = circle − minor segment.
- Shaded area = **total area − unshaded parts** (or sum of parts).
- Wheel problems: **distance = number of turns x circumference**.

CHAPTER 12 · MATHS

Surface Areas and Volumes

A medicine capsule, an ice-cream cone with a scoop, a tent made of a cylinder topped by a cone and a metal ball melted into wire are all solids made of simpler ones. This chapter combines the formulae for the cylinder, cone, sphere and hemisphere to find surface areas and volumes of combined solids and to solve problems where a solid is melted and recast into another shape.

Big Questions

- What are the formulae for the basic solids?
- How do we find the surface area and volume of a combination of solids?
- Why does the volume stay the same when a solid is melted and recast?

12.1 Formulae of Basic Solids

Solid	Curved surface area (CSA)	Total surface area (TSA)	Volume
Cylinder (r, h)	$2\pi rh$	$2\pi r(r + h)$	$\pi r^2 h$
Cone (r, h, slant l)	πrl	$\pi r(l + r)$	$(1/3)\pi r^2 h$
Sphere (r)	$4\pi r^2$	$4\pi r^2$	$(4/3)\pi r^3$
Hemisphere (r)	$2\pi r^2$	$3\pi r^2$	$(2/3)\pi r^3$
Cuboid (l, b, h)	$2h(l + b)$	$2(lb + bh + hl)$	lbh
Cube (a)	$4a^2$	$6a^2$	a^3

FORMULA

Slant height of a cone: $l = \sqrt{(r^2 + h^2)}$. Cone: radius 7, height 24, so l = 25, CSA = (22/7)(7)(25) = **550**, TSA = 550 + 154 = **704**, volume = (1/3)(22/7)(49)(24) = **1232**. Cylinder (7, 10): volume = 1540, TSA = 2(22/7)(7)(17) = **748**.

12.2 Combination of Solids

IMPORTANT

When two solids are joined, the **surfaces that meet become hidden**, so the **total surface area of the combination is the sum of the visible curved (or flat) areas**, **not** the sum of the TSAs. **Volume** of the combination is simply the **sum of the volumes**.

Combination	Visible surface area	Volume
Cone on a hemisphere (toy, top)	CSA of cone + CSA of hemisphere	Cone + hemisphere
Cylinder with hemispherical ends (capsule)	CSA of cylinder + 2 hemispherical CSAs	Cylinder + sphere of same radius

Combination	Visible surface area	Volume
Cylinder with a cone on top (tent, rocket)	CSA cylinder + CSA cone + base if closed below	Cylinder + cone
Cylinder with a hemispherical hollow at one end	CSA of cylinder + CSA of hemisphere + remaining ring/base area	Cylinder − hemisphere
Cube with a hemispherical depression	TSA of cube − πr^2 + $2\pi r^2$	Cube − hemisphere

Example (toy). A cone of radius 7 cm and height 24 cm stands on a hemisphere of radius 7 cm. Surface area = cone CSA 550 + hemisphere CSA 2(22/7)(49) = 308, total **858 cm^2**. Volume = 1232 + (2/3)(22/7)(343) = 1232 + 718.67 = **1950.67 cm^3**.

Example (capsule). A capsule has a cylinder of length 5 mm and radius 2 mm with hemispherical ends. Surface area = $2\pi rh + 2(2\pi r^2) = 2\pi(2)(5) + 4\pi(4) = 20\pi + 16\pi = 36\pi$ = **113.1 mm^2**. Volume = $\pi(4)(5) + (4/3)\pi(8) = 20\pi + 32\pi/3$ = **96.3 mm^3**.

Example (wooden cylinder with a hemispherical scoop at each end). TSA = CSA of cylinder + 2 x CSA of hemisphere, as the flat circular faces are removed and replaced by curved faces.

12.3 Conversion of a Solid into Another

FORMULA

When a solid is melted and recast, **volume remains the same**: volume of the original = volume of the new solid(s). If n identical pieces are made, **n x (volume of one) = volume of the original**.

Problem	Working	Answer
A metal sphere of radius 4.2 cm is melted into a cylinder of radius 6 cm	$(4/3)(4.2)^3$ = 36 h, 98.784 = 36h	h = **2.744 cm**
A cylinder of radius 3 cm and height 8 cm is melted into spheres of radius 1 cm	$72\pi / (4\pi/3)$ = 54	**54 spheres**
Sphere of radius 3 cm melted into 27 identical small spheres	$36\pi = 27(4/3)\pi r^3$ gives r^3 = 1	r = **1 cm**
Water flowing through a pipe of radius 1 cm at 5 m/s fills a tank in a given time	Volume per second = π x 1 x 500 cm^3	about 1571 cm^3 (1.57 litres) every second

Earth work. A well of diameter 7 m dug 20 m deep has soil spread evenly over a field 22 m by 14 m; volume = $(22/7)(3.5)^2(20)$ = 770 m^3; height of the platform = 770/(22 x 14) = **2.5 m**.

Cone and cylinder. A cone of radius r and height h is cast into a cylinder of equal radius: the height of the cylinder is h/3.

REMEMBER

Check the units of all dimensions before using the formulae (convert cm to m, litres to cm^3: 1 litre = 1000 cm^3).

Quick Revision: Likely Exam Questions

Likely question	Model answer
Volume of a hemisphere of radius 3.5 cm?	(2/3)(22/7)(42.875) = 89.83 cm^3.
Slant height of a cone with r = 6, h = 8?	10.
Why is TSA of a combination not the sum of TSAs?	The joined faces are hidden.
A sphere's radius is doubled; by what factor does volume change?	8.
Ratio of the volume of a cone and a cylinder of same base and height?	1 : 3.
Total surface area of a hemisphere of radius 7 cm?	$3\pi r^2$ = 462 cm^2.
When a solid is melted and recast, what stays the same?	Volume.
Formula for the capsule's surface area?	$2\pi rh + 4\pi r^2$.

★ KEY TAKEAWAYS

- Learn the **CSA, TSA and volume** formulae for cylinder, cone, sphere, hemisphere.
- $\mathbf{l^2 = r^2 + h^2}$ for a cone.
- Combination: **add volumes; add visible surface areas only**.
- Recasting: **volume before = volume after**; count pieces by dividing volumes.

CHAPTER 13 · MATHS

Statistics

A school wants to know the typical marks of its Class 10, a doctor the usual pulse rate of patients and a factory the usual lifetime of a bulb. When the data are collected into class intervals, a single representative value, a measure of central tendency, summarises them. This chapter finds the mean, median and mode of grouped data and relates the three.

Big Questions

- How do we find the mean of data given in class intervals?
- How are median and modal class located and the values calculated?
- How are mean, median and mode connected?

13.1 Mean of Grouped Data

DEFINITION

In a frequency distribution, each class is represented by its **class mark** x_i = (upper limit + lower limit)/2. The **mean** is the sum of f_ix_i divided by the sum of the frequencies.

FORMULA

Direct method: mean = $\Sigma f_ix_i / \Sigma f_i$. **Assumed mean method:** mean = $a + \Sigma f_id_i / \Sigma f_i$, where a is an assumed mean (usually the class mark near the middle) and $\mathbf{d_i = x_i - a}$.

Class	f_i	Class mark x_i	f_ix_i	d_i (a = 35)	f_id_i
10–20	5	15	75	−20	−100
20–30	8	25	200	−10	−80
30–40	12	35	420	0	0
40–50	10	45	450	10	100
50–60	5	55	275	20	100
Total	**40**		**1420**		**20**

Direct method: mean = 1420/40 = **35.5**. **Assumed mean method:** mean = 35 + 20/40 = **35.5**. Both methods agree; the second keeps the numbers small.

13.2 Mode of Grouped Data

FORMULA

The **modal class** is the class with the **highest frequency**. **Mode = $l + [(f_1 - f_0) / (2f_1 - f_0 - f_2)] \times h$**, where l = lower limit of the modal class, h = class size, f_1 = frequency of the modal class, f_0 = frequency of the class before it, f_2 = frequency of the class after it.

From the table above: modal class 30–40, l = 30, h = 10, f_1 = 12, f_0 = 8, f_2 = 10: mode = 30 + (4/(24 − 18)) x 10 = 30 + 6.67 = **36.67**.

13.3 Median of Grouped Data

Step 1 — Make a cumulative frequency (cf) column
Add frequencies progressively; the last cf equals n.

Step 2 — Find n/2 and the median class
The median class is the class whose cumulative frequency is **just greater than or equal to n/2**.

Step 3 — Apply the formula
Use l, cf, f and h for the median class.

FORMULA

Median = l + [(n/2 − cf) / f] x h, where l = lower limit of the median class, n = total frequency, cf = cumulative frequency of the class **before** the median class, f = frequency of the median class, h = class size.

Class	f	cf
10–20	5	5
20–30	8	13
30–40	12	25
40–50	10	35
50–60	5	40

Here n = 40, n/2 = 20, so the median class is 30–40 (cf = 25 ≥ 20). l = 30, cf (before) = 13, f = 12, h = 10: median = 30 + (7/12) x 10 = **35.83**.

13.4 Relationship between the Three

FORMULA

3 Median = Mode + 2 Mean (an empirical relation for moderately skewed data). Check: 3 x 35.83 = 107.5 and 36.67 + 2 x 35.5 = 107.67, which are nearly equal.

Measure	Meaning	Use
Mean	The average of all observations	Marks, wages, when no extreme values
Median	The middle value; half the observations lie below it	Income, where a few values are very large
Mode	The most frequent value	Most common shoe size, most popular item

13.5 Worked Examples

Missing frequency. Classes 0–10, 10–20, 20–30 have frequencies 4, x and 6 and the mean is 16. Class marks are 5, 15, 25, so (20 + 15x + 150)/(10 + x) = 16, giving 170 + 15x = 160 + 16x and **x = 10**.

Finding the mode. Classes 0–20, 20–40, 40–60, 60–80 with frequencies 4, 10, 6, 5: modal class 20–40; mode = 20 + (6/(20 − 4 − 6)) x 20 = 20 + 12 = **32**.

Finding the median. Frequencies 3, 7, 12, 10, 8 (n = 40) for classes of width 10 starting at 0: cf = 3, 10, 22, 32, 40; n/2 = 20, median class 20–30; median = 20 + (20 − 10)/12 x 10 = **28.33**.

> **IMPORTANT**
> Classes must be **continuous (exclusive)** for the formulae; if the classes are written 10–19, 20–29 convert them first. The classes must also have the same width for the mode formula.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is a class mark?	The mid-value of a class: (upper + lower)/2.
Write the formula of the mean by the assumed mean method.	$a + \Sigma f_i d_i / \Sigma f_i$.
What is the modal class?	The class with the highest frequency.
How do you find the median class?	Locate n/2 in the cumulative frequency column.
If mean = 12 and median = 11.5, estimate the mode.	3(11.5) − 2(12) = 10.5.
What does cf mean in the median formula?	Cumulative frequency of the class preceding the median class.
When is median preferred?	When the data have extreme values.
Find the class mark of 25–35.	30.

★ KEY TAKEAWAYS

- Mean = **Σfx/Σf**, or **a + Σfd/Σf**.
- Mode = **$l + (f_1 - f_0)/(2f_1 - f_0 - f_2) \times h$**.
- Median = **$l + (n/2 - cf)/f \times h$**.
- **3 Median = Mode + 2 Mean.**

CHAPTER 14 · MATHS

Probability

Will it rain tomorrow? Which team will win the toss? Questions about uncertain events can be answered with numbers called probabilities. This chapter defines theoretical (classical) probability for experiments whose outcomes are equally likely and applies it to coins, dice, cards and everyday draws.

Big Questions

- What is the probability of an event and what values can it take?
- How are an event and its complement connected?
- How do we count outcomes for dice, cards and bags?

14.1 Basic Terms

Term	Meaning	Example
Experiment	An action with several possible results	Tossing a coin
Outcome	A possible result	Head or tail
Sample space	The set of all possible outcomes	{H, T}
Event	A collection of outcomes	Getting an even number on a die
Elementary event	An event with a single outcome	Getting a 6
Equally likely	Each outcome has the same chance	Fair coin, unbiased die

14.2 Theoretical Probability

FORMULA

P(E) = (number of outcomes favourable to E) / (total number of possible outcomes), when all outcomes are equally likely. **0 ≤ P(E) ≤ 1.** A **sure (certain)** event has probability **1**; an **impossible** event has probability **0**. The sum of the probabilities of all elementary events of an experiment is **1**.

FORMULA

Complementary events: P(E) + P(not E) = 1, so **P(not E) = 1 − P(E)**. The 'not E' event is written E-bar.

Examples. A coin: P(head) = 1/2. A die: P(getting 4) = 1/6; P(an even number) = 3/6 = 1/2; P(a prime number) = 3/6 (2, 3, 5); P(a number less than 7) = 1; P(a number greater than 6) = 0. A bag with 3 red and 5 blue balls: P(red) = 3/8 and P(not red) = 5/8. P(a pen is good) with 20 defective in 144 pens = 124/144 = **31/36**.

14.3 Two Coins and Two Dice

Experiment	Sample space	Event and probability
Two coins	HH, HT, TH, TT (4)	P(two heads) = 1/4; P(at least one head) = 3/4; P(exactly one head) = 1/2

Experiment	Sample space	Event and probability
Three coins	8 outcomes	P(at least one tail) = 7/8; P(all heads) = 1/8
Two dice	6 x 6 = **36** outcomes	P(sum 8) = 5/36; P(sum 7) = 6/36 = 1/6; P(doublet) = 6/36 = 1/6; P(sum 2 or 12) = 2/36 = 1/18

Sum on two dice	Favourable outcomes	Number
2	(1,1)	1
3	(1,2),(2,1)	2
4	(1,3),(2,2),(3,1)	3
5	4 pairs	4
6	5 pairs	5
7	6 pairs	6
8	(2,6),(3,5),(4,4),(5,3),(6,2)	5
9	4 pairs	4
10	3 pairs	3
11	2 pairs	2
12	(6,6)	1

REMEMBER

On two dice **sum = 7 is the most likely** (6 ways); the totals of the 11 sums are 36. P(sum < 4) = 3/36, P(sum > 10) = 3/36, P(product 6) = 4/36 = 1/9 (outcomes 1,6; 2,3; 3,2; 6,1).

14.4 Playing Cards

A pack has **52 cards** in 4 suits (spades and clubs: black; hearts and diamonds: red) of 13 cards each: A, 2 to 10, J, Q, K. There are **12 face cards** (J, Q, K in each suit) and 4 aces.

Event	Favourable	Probability
A king	4	1/13
The king of hearts	1	1/52
A red card	26	1/2
A red face card	6	3/26
A face card	12	3/13
A spade	13	1/4
Not an ace	48	12/13
The queen of diamonds or a jack	1 + 4 − 0	5/52

IMPORTANT

Do not count the same outcome twice: in 'a king or a heart', the king of hearts is common, so favourable = 4 + 13 − 1 = 16 and P = 16/52 = 4/13. In a die question, always write the sample space when in doubt.

14.5 Worked Examples

Lot of numbered tickets. Tickets numbered 1 to 20 are shuffled; P(multiple of 3 or 5): multiples of 3: 3, 6, 9, 12, 15, 18 (6); of 5: 5, 10, 15, 20 (4); 15 is common, so 6 + 4 − 1 = 9 and P = **9/20**.

Spinner/bag. A bag has 4 red, 5 white and 6 black balls: total 15; P(white) = 1/3; P(not black) = 9/15 = 3/5; P(red or white) = 9/15 = 3/5.

Unknown count. A bag has 5 red balls and some blue balls. If P(red) = 1/3, the total is 5 x 3 = 15, so there are **10 blue balls**. If one red ball is removed, P(red) becomes 4/14 = 2/7.

Leap year. A year of 366 days has 52 complete weeks and 2 extra days; the chance of exactly 53 Sundays is 2/7.

Quick Revision: Likely Exam Questions

Likely question	Model answer
What is the probability of a sure event?	1.
Can a probability be 1.3?	No; it lies between 0 and 1.
If P(E) = 0.05, find P(not E).	0.95.
A die is thrown once; P(number > 4)?	2/6 = 1/3.
Two coins are tossed; P(no head)?	1/4.
A card is drawn from a pack; P(a black king)?	2/52 = 1/26.
Two dice are thrown; P(sum is 9)?	4/36 = 1/9.
Why must the outcomes be equally likely?	The formula counts outcomes equally; an unfair coin breaks this.

★ **KEY TAKEAWAYS**

- **P(E) = favourable / total**, outcomes equally likely.
- **0 ≤ P(E) ≤ 1**; sure = 1, impossible = 0.
- **P(E) + P(not E) = 1.**
- Two dice: **36** outcomes; cards: **52**; do not double count common outcomes.